Band-1 Hadamard order 13908 – Classical Paley-I (q=13907); IN SPK via Paley-I; Gap 2052 Still OPEN

H_13908 Paley-I PASS; Paley-II FAIL/SKIP (q=6953=17*409 NOT prime; ===1 mod 4 but composite); Kronecker FAIL/SKIP (4*3477; 3477===1 mod 4 NOT Hadamard-admissible; 3477=3*19*61 (odd composite)); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-07

Document class. Scientific computational / construction certificate (Band-1).

Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.

Parents. `2026-10-07_Band1_Hadamard_13904_PaleyI`; `2026-10-07_Band1_Hadamard_13884_PaleyI`; `2026-10-07_Band1_Hadamard_13880_PaleyI`; `2026-10-07_Band1_Hadamard_13860_PaleyI`; `2026-10-07_Band1_Hadamard_13832_PaleyI`; `2026-10-07_Band1_Hadamard_13808_PaleyI`; `2026-10-07_Band1_Hadamard_13800_PaleyI`; `2026-10-07_Band1_Hadamard_13764_PaleyI`; `2026-10-07_Band1_Hadamard_13760_PaleyI`; `2026-10-07_Band1_Hadamard_13752_PaleyI`; `2026-10-07_Band1_Hadamard_13724_PaleyI`; `2026-10-07_Band1_Hadamard_13712_PaleyI`; `2026-10-07_Band1_Hadamard_13692_PaleyI`; `2026-10-07_Band1_Hadamard_13688_PaleyI`; `2026-10-06_Band1_Hadamard_13680_PaleyI`; `2026-10-06_Band1_Hadamard_13628_PaleyI`; `2026-10-06_Band1_Hadamard_13620_PaleyI`; `2026-10-06_Band1_Hadamard_13592_PaleyI`; `2026-10-06_Band1_Hadamard_13568_PaleyI`; `2026-10-06_Band1_Hadamard_13524_PaleyI`; `2026-10-06_Band1_Hadamard_13500_PaleyI`; `2026-10-06_Band1_Hadamard_13488_PaleyI`; `2026-10-06_Band1_Hadamard_13464_PaleyI`; `2026-10-06_Band1_Hadamard_13452_PaleyI`; `2026-10-06_Band1_Hadamard_13412_PaleyI`; `2026-10-06_Band1_Hadamard_13400_PaleyI`; `2026-10-06_Band1_Hadamard_13368_PaleyI`; `2026-10-06_Band1_Hadamard_13340_PaleyI`; `2026-10-06_Band1_Hadamard_13332_PaleyI`; `2026-10-06_Band1_Hadamard_13328_PaleyI`; `2026-10-06_Band1_Hadamard_13292_PaleyI`; `2026-10-06_Band1_Hadamard_13268_PaleyI`; `2026-10-06_Band1_Hadamard_13260_PaleyI`; `2026-10-06_Band1_Hadamard_13220_PaleyI`; `2026-10-06_Band1_Hadamard_13188_PaleyI`; `2026-10-06_Band1_Hadamard_13184_PaleyI`; `2026-10-06_Band1_Hadamard_13172_PaleyI`; `2026-10-05_Band1_Hadamard_13164_PaleyI`; `2026-10-05_Band1_Hadamard_13160_PaleyI`; `2026-10-05_Band1_Hadamard_13152_PaleyI`; `2026-10-05_Band1_Hadamard_13148_PaleyI`; `2026-10-05_Band1_Hadamard_13128_PaleyI`; `2026-10-05_Band1_Hadamard_13104_PaleyI`; `2026-10-05_Band1_Hadamard_13100_PaleyI`; `2026-10-05_Band1_Hadamard_13064_PaleyI`; `2026-10-05_Band1_Hadamard_13044_PaleyI`; `2026-10-05_Band1_Hadamard_13008_PaleyI`; `2026-10-05_Band1_Hadamard_13004_PaleyI`;

2026-10-05_Band1_Hadamard_12984_PaleyI; 2026-10-05_Band1_Hadamard_12980_PaleyI;
2026-10-05_Band1_Hadamard_12968_PaleyI; 2026-10-05_Band1_Hadamard_12960_PaleyI;
2026-10-05_Band1_Hadamard_12924_PaleyI; 2026-10-05_Band1_Hadamard_12920_PaleyI;
2026-10-05_Band1_Hadamard_12912_PaleyI; 2026-10-05_Band1_Hadamard_12908_PaleyI;
2026-10-04_Band1_Hadamard_12900_PaleyI; 2026-10-04_Band1_Hadamard_12824_PaleyI;
2026-10-04_Band1_Hadamard_12800_PaleyI; 2026-10-04_Band1_Hadamard_12792_PaleyI;
2026-10-04_Band1_Hadamard_12764_PaleyI; 2026-10-04_Band1_Hadamard_12744_PaleyI;
2026-10-04_Band1_Hadamard_12740_PaleyI; 2026-10-04_Band1_Hadamard_12704_PaleyI;
2026-10-04_Band1_Hadamard_12672_PaleyI; 2026-10-04_Band1_Hadamard_12660_PaleyI;
2026-10-04_Band1_Hadamard_12648_PaleyI; 2026-10-04_Band1_Hadamard_12620_PaleyI;
2026-10-04_Band1_Hadamard_12612_PaleyI; 2026-10-04_Band1_Hadamard_12584_PaleyI;
2026-10-04_Band1_Hadamard_12548_PaleyI; 2026-10-04_Band1_Hadamard_12540_PaleyI;
2026-10-03_Band1_Hadamard_12380_PaleyI; 2026-10-03_Band1_Hadamard_12348_PaleyI;
2026-10-03_Band1_Hadamard_12344_PaleyI; 2026-10-03_Band1_Hadamard_12324_PaleyI;
2026-10-03_Band1_Hadamard_12264_PaleyI; 2026-10-03_Band1_Hadamard_12252_PaleyI;
2026-10-03_Band1_Hadamard_12240_PaleyI; 2026-10-03_Band1_Hadamard_12228_PaleyI;
2026-10-03_Band1_Hadamard_12212_PaleyI; 2026-10-03_Band1_Hadamard_12204_PaleyI;
2026-10-03_Band1_Hadamard_12164_PaleyI; 2026-10-03_Band1_Hadamard_12144_PaleyI;
2026-10-03_Band1_Hadamard_12120_PaleyI; 2026-10-03_Band1_Hadamard_12108_PaleyI;
2026-10-03_Band1_Hadamard_12072_PaleyI; 2026-10-03_Band1_Hadamard_12044_PaleyI;
2026-10-03_Band1_Hadamard_12012_PaleyI; 2026-10-03_Band1_Hadamard_12008_PaleyI;
2026-10-03_Band1_Hadamard_11988_PaleyI; 2026-10-03_Band1_Hadamard_11972_PaleyI;
2026-10-03_Band1_Hadamard_11960_PaleyI; 2026-10-03_Band1_Hadamard_11940_PaleyI;
2026-10-03_Band1_Hadamard_11928_PaleyI; 2026-10-02_Band1_Hadamard_11924_PaleyI;
2026-10-02_Band1_Hadamard_11904_PaleyI; 2026-10-02_Band1_Hadamard_11888_PaleyI;
2026-10-02_Band1_Hadamard_11868_PaleyI; 2026-10-02_Band1_Hadamard_11864_PaleyI;
2026-10-02_Band1_Hadamard_11840_PaleyI; 2026-10-02_Band1_Hadamard_11832_PaleyI;
2026-10-02_Band1_Hadamard_11828_PaleyI; 2026-10-02_Band1_Hadamard_11808_PaleyI;
2026-10-02_Band1_Hadamard_11784_PaleyI; 2026-10-02_Band1_Hadamard_11780_PaleyI;
2026-10-02_Band1_Hadamard_11744_PaleyI; 2026-10-02_Band1_Hadamard_11732_PaleyI;
2026-10-02_Band1_Hadamard_11720_PaleyI; 2026-10-02_Band1_Hadamard_11700_PaleyI;
2026-09-30_Band1_Hadamard_11588_PaleyI; 2026-09-30_Band1_Hadamard_11580_PaleyI;
2026-09-30_Band1_Hadamard_11552_PaleyI; 2026-09-30_Band1_Hadamard_11528_PaleyI;
2026-09-30_Band1_Hadamard_11520_PaleyI; 2026-09-30_Band1_Hadamard_11504_PaleyI;
2026-09-30_Band1_Hadamard_11492_PaleyI; 2026-09-30_Band1_Hadamard_11484_PaleyI;
2026-09-30_Band1_Hadamard_11472_PaleyI; 2026-09-30_Band1_Hadamard_11468_PaleyI;
2026-09-30_Band1_Hadamard_11448_PaleyI; 2026-09-30_Band1_Hadamard_11444_PaleyI;
2026-09-30_Band1_Hadamard_11424_PaleyI; 2026-09-30_Band1_Hadamard_11412_PaleyI;
2026-09-30_Band1_Hadamard_11400_PaleyI; 2026-09-30_Band1_Hadamard_11384_PaleyI;
2026-09-30_Band1_Hadamard_11352_PaleyI; 2026-09-30_Band1_Hadamard_11312_PaleyI;
2026-09-30_Band1_Hadamard_11300_PaleyI; 2026-09-30_Band1_Hadamard_11288_PaleyI;

2026-09-30_Band1_Hadamard_11280_PaleyI; 2026-09-30_Band1_Hadamard_11252_PaleyI;
2026-09-29_Band1_Hadamard_11244_PaleyI; 2026-09-29_Band1_Hadamard_11240_PaleyI;
2026-09-29_Band1_Hadamard_11172_PaleyI; 2026-09-29_Band1_Hadamard_11160_PaleyI;
2026-09-29_Band1_Hadamard_11132_PaleyI; 2026-09-29_Band1_Hadamard_11120_PaleyI;
2026-09-29_Band1_Hadamard_11088_PaleyI; 2026-09-29_Band1_Hadamard_11084_PaleyI;
2026-09-29_Band1_Hadamard_11072_PaleyI; 2026-09-29_Band1_Hadamard_11060_PaleyI;
2026-09-29_Band1_Hadamard_11048_PaleyI; 2026-09-29_Band1_Hadamard_11028_PaleyI;
2026-09-29_Band1_Hadamard_11004_PaleyI; 2026-09-29_Band1_Hadamard_10988_PaleyI;
2026-09-29_Band1_Hadamard_10980_PaleyI; 2026-09-29_Band1_Hadamard_10940_PaleyI;
2026-09-29_Band1_Hadamard_10892_PaleyI; 2026-09-29_Band1_Hadamard_10884_PaleyI;
2026-09-29_Band1_Hadamard_10860_PaleyI; 2026-09-29_Band1_Hadamard_10848_PaleyI;
2026-09-29_Band1_Hadamard_10832_PaleyI; 2026-09-29_Band1_Hadamard_10800_PaleyI;
2026-09-29_Band1_Hadamard_10772_PaleyI; 2026-09-29_Band1_Hadamard_10740_PaleyI;
2026-09-29_Band1_Hadamard_10724_PaleyI; 2026-09-29_Band1_Hadamard_10712_PaleyI;
2026-09-28_Band1_Hadamard_10692_PaleyI; 2026-09-28_Band1_Hadamard_10688_PaleyI;
2026-09-28_Band1_Hadamard_10668_PaleyI; 2026-09-28_Band1_Hadamard_10652_PaleyI;
2026-09-28_Band1_Hadamard_10640_PaleyI; 2026-09-28_Band1_Hadamard_10632_PaleyI;
2026-09-28_Band1_Hadamard_10628_PaleyI; 2026-09-28_Band1_Hadamard_10608_PaleyI;
2026-09-28_Band1_Hadamard_10568_PaleyI; 2026-09-28_Band1_Hadamard_10560_PaleyI;
2026-09-28_Band1_Hadamard_10532_PaleyI; 2026-09-28_Band1_Hadamard_10500_PaleyI;
2026-09-28_Band1_Hadamard_10488_PaleyI; 2026-09-28_Band1_Hadamard_10464_PaleyI;
2026-09-28_Band1_Hadamard_10460_PaleyI; 2026-09-28_Band1_Hadamard_10428_PaleyI;
2026-09-28_Band1_Hadamard_10400_PaleyI; 2026-09-24_Band1_Hadamard_10392_PaleyI;
2026-09-24_Band1_Hadamard_10344_PaleyI; 2026-09-24_Band1_Hadamard_10332_PaleyI;
2026-09-24_Band1_Hadamard_10304_PaleyI; 2026-09-24_Band1_Hadamard_10272_PaleyI;
2026-09-23_Band1_Hadamard_10268_PaleyI; 2026-09-23_Band1_Hadamard_10224_PaleyI;
2026-09-23_Band1_Hadamard_10212_PaleyI; 2026-09-23_Band1_Hadamard_10164_PaleyI;
2026-09-23_Band1_Hadamard_10160_PaleyI; 2026-09-23_Band1_Hadamard_10152_PaleyI;
2026-09-23_Band1_Hadamard_10140_PaleyI; 2026-09-23_Band1_Hadamard_10112_PaleyI;
2026-09-23_Band1_Hadamard_10104_PaleyI; 2026-09-23_Band1_Hadamard_10100_PaleyI;
2026-09-23_Band1_Hadamard_10068_PaleyI.

Lab. `/workspace/experiments/band1/hadamard_13908_paleyI/`.

Cutoff. 2026-10-07T23:30:44Z (~7:30PM ET).

Honesty stamp (non-negotiable).

This memo claims **finite construction of order 13908 only (verified max|HHˆT-nI|=0), via classical Paley-I q=13907 (prime ===3 mod 4). 13908 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 13908: Paley-I **PASS** ($q = 13907$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=6953=17*409 NOT prime; ===1 mod 4 but composite); Kronecker **FAIL/SKIP** (note $13908 = 4 \cdot 3477$; $3477 \equiv 1 \pmod 4$ NOT Hadamard-admissible; $3477 = 3 \cdot 19 \cdot 61$ (odd composite)). **13908** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard

conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **13904+13884+13880+13860+13832+13808+13800+13764+13760+13752+13724+13712+13692+136** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

ewpage

Abstract

Finite construction of order **13908**.

- H_{13908} = Paley-I ($q = 13907$) – classical SPK
- Paley-II for 13908: **FAIL/SKIP** ($q = 6953 = 17 \cdot 409$ NOT prime; $\equiv 1 \pmod 4$ but composite)
- Kronecker for 13908: **FAIL/SKIP** ($4 \cdot 3477$; $3477 \equiv 1 \pmod 4$ NOT Hadamard-admissible; $3477 = 3 \cdot 19 \cdot 61$ (odd composite))
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_13908.py`

Results. overall_pass=True; units 620/620; row blocks 109/109 (block size 128); wall 2560.8071s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-07T23:30:44Z (~7:30PM ET)

Arithmetic

- Target $n = 13908 = q + 1$ with $q = 13907$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 6953 = 17 \cdot 409$ NOT prime; $\equiv 1 \pmod 4$ but composite — FAIL/SKIP.
- Kronecker: $13908 = 4 \cdot 3477$; $3477 \equiv 1 \pmod 4$ NOT Hadamard-admissible; $3477 = 3 \cdot 19 \cdot 61$ (odd composite) — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_13908 Paley-I	Paley-I q=13907	0	2521.2082
H_13908 Paley-II	FAIL/SKIP (q=6953=17*409 NOT prime; ===1 mod 4 but composite)	–	–

Object	Method	max|HHˆT-nI|	Wall (s)
H_13908 Kronecker	FAIL/SKIP (4*3477; 3477===1 mod 4 NOT Hadamard-admissible; 3477=3*19*61 (odd composite))	–	–

SPK closure note

Seeds include prior SPK through **11528** lineage (**11528**; **11520**; **11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**; **11252**; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; …; **NOT** 8948/8644/8516/…) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

Quantity	Value
12488 in SPK	**True** (Paley-I q=13907)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims

- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q2=6961->n=13924 Paley-II DIRECT (first n===0 mod 4 after 13908 with a classical Paley route; q2=6961 prime ===1 mod 4; Paley-I q=13923=3^2*71317 *NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59^2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13912/13916/13920 have NO classical Paley route (13920: q2=6959 prime but ===3 mod 4 needs ===1); 13924=2^2*59^2 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII) – the Paley-I-only chain has skipped such orders, e.g. 13900) | next Paley-I DIRECT: q=13931->n=13932 (prime ===3 mod 4; 13913, 13921 primes ===1 mod 4 skipped; no prime ===3 mod 4 in 13908..13930; Paley-II q=6965=5*7*199 NOT prime (not a prime power); ===1 mod 4 but composite -> FAIL/SKIP; Kronecker 13932=4*3483; 3483=3?*43 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3483); 13932=2^2*3?*43 ===12 mod 16 = 13884-like odd-m shape (q2 composite ===1, m odd ===3 mod 4)) (or gap 2052 / Track B)